14 An ideal gas has thermodynamic temperature T, and its molecules has root-mean-square speed c_{rms}. If the gas is heated to temperature $2T$, what is the new root-mean-square speed of the molecules?

A $\sqrt{2}c_{rms}$ **B** $2c_{rms}$ **C** $2\sqrt{2}c_{rms}$ **D** $4c_{rms}$